

When Form Outlives State: Distinguishing Extended Phenotypic Expression from Zombie Outputs in Large Language Models

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Abstract

Large language models can reproduce culturally recognizable forms with remarkable fidelity. Whether those outputs express internal states or merely reconstruct surface regularities remains a different question. Recent activation-injection experiments provide causal evidence that some models can, under restricted conditions, detect and report selected internal representations. This paper uses that result to sharpen Extended Phenotype Theory as applied to LLMs. I distinguish four state-output relations: introspectively accessible extended phenotype, opaque extended phenotype, phenotypic residue, and state-expression decoupling. The third category captures what I call a Zombie output: a culturally selected form that survives in the model's behavior without demonstrated access to the cognitive or normative structure historically associated with it. I then propose a causal protocol combining representation identification, activation intervention, output measurement, privileged-access controls, and adversarial stability tests. The protocol does not infer consciousness, subjective experience, or legal intention. Its narrower purpose is to determine when output regularity carries evidence of a relevant internal state and when it does not. The resulting framework turns a broad evolutionary analogy into a falsifiable research programme and supplies an institutional rule: LLM self-reports should not support reliance unless their state-output relation survives causal, comparative, and strategic testing.

Keywords: extended phenotype theory; large language models; introspection; activation steering; cultural evolution; Zombie outputs; AI governance; representation engineering

1. Introduction

A legal maxim can appear in a brief without doing the work that once made the maxim authoritative. A management slogan can appear in a strategic recommendation without answering the company's actual problem. A model can say that it is thinking about a concept without having inspected any internal state that corresponds to the report. In each case the form survives. What remains uncertain is whether the state, mechanism, or cognitive structure associated with that form survives with it.

That uncertainty matters because large language models are increasingly asked to explain their reasoning, identify their biases, disclose uncertainty, and distinguish intended from unintended outputs. A fluent self-report invites the user to treat linguistic coherence as evidence of privileged access. Yet conversation alone cannot distinguish grounded self-report from a plausible continuation learned from data. The output may be accurate by coincidence, derived from public facts about the model, inferred from prior tokens, or confabulated in response to the social demands of the prompt.

Lindsey (2026) advances this debate by intervening on model activations rather than relying on conversation alone. Concept vectors are injected into selected layers, after which models are asked to identify or reason about the resulting states. Some models sometimes detect the injected representation, distinguish it from text input, relate prior activation to an apparently unintended output, and modulate internal representations when instructed or incentivized to think about a concept. The effects are unreliable, sensitive to layer, strength, prompt, model, and post-training, and particularly visible in Claude Opus 4 and 4.1. Failures remain common. The experiments do not establish phenomenal consciousness or a general faculty of self-knowledge. They do establish something narrower and methodologically important: for selected cases, a causal relation can be traced from an internal representation to a model's report about that representation.

This paper asks what that result changes for an evolutionary account of LLM outputs. Earlier work proposed that systematic LLM behaviors can be understood as the Extended Phenotype of LLMs, or EPL: outputs instantiate patterns selected through the training corpus and post-training environment rather than merely reflecting neutral information (Lerer, 2026a). A second paper identified the inverse risk in legal language: a model may reproduce the form of a legal meme without activating the cognitive and intentional structure that historically gave the form its function, producing what that paper called Zombie Law (Lerer, 2026b). These propositions are compatible, but they are not identical. One describes expression. The other describes survival of form after functional depth has become uncertain.

The distinction has not yet been operationalized. Observable output alone cannot show whether a relevant representation exists internally, whether the system has privileged access to it, or whether the output is causally dependent on it. The central claim of this paper is therefore methodological: **extended phenotypic expression in an LLM should be classified by the causal relation between internal representation and external output, not by output resemblance alone.**

I develop that claim in four steps. Section 2 separates cultural inheritance, internal representation, and output expression. Section 3 places current work on LLM introspection within that distinction. Section 4 proposes a four-state taxonomy. Section 5 presents an empirical protocol and falsifiable predictions. Section 6 explains the multilevel evolutionary dynamics. Section 7 derives a rule for institutional reliance. The argument remains deliberately bounded. It does not claim that LLMs possess human-like understanding, consciousness, or legal intention. It asks a prior question: when does the form carry evidence of the state?

2. From Cultural Selection to Model Output

2.1 The extended phenotype claim

Dawkins's extended phenotype thesis directs analysis away from the boundary of the organism and toward the effects through which replicators alter their environment (Dawkins, 1982). The familiar examples are material, such as dams, webs, and host manipulation. In cultural evolution, the analogous move is not to declare every artifact a phenotype. It is to investigate whether cultural variants generate external effects that increase their retention, transmission, or capacity to structure later selection (Campbell, 1974; Hull, 1988; Dennett, 1995).

LLMs introduce a distinctive transmission environment. Human-produced texts are selected into datasets. Training compresses regularities from those texts into model parameters and representations. Post-training adds further selection through preference data, reward functions, policy constraints, and behavioral evaluation. Deployment then externalizes selected regularities as text and action. Those outputs re-enter human institutions and, increasingly, future training corpora. The resulting loop is compatible with an extended-phenotype analysis because external expressions modify the environment in which later expressions are judged, retained, and copied.

Compatibility is not confirmation. A model output can resemble a culturally successful form for several reasons. The pattern may be represented internally and causally drive the output. It may be stored but inaccessible to the model's self-report mechanisms. It may arise from shallow token associations without a representation corresponding to the interpretation supplied by an observer. Or a relevant state may be present while the output is distorted by prompting, post-training, or strategic behavior. A useful EPL account must distinguish these possibilities.

2.2 Three stages that should not be collapsed

The first stage is **selection history**. A concept, phrase, norm, or rhetorical structure becomes overrepresented because it possesses some combination of frequency, prestige, affective valence, institutional support, or compatibility with the objectives used in training. Selection history explains why a pattern was available to be learned. It does not prove how a particular model represents it.

The second stage is **internal realization**. Training produces distributed representations and mechanisms that may track concepts, features, relations, or behavioral dispositions. Representation engineering can identify and manipulate some population-level directions, but an interpretable vector is not automatically a complete semantic object (Zou et al., 2023). Different internal mechanisms may yield similar outputs, while similar mechanisms may be elicited differently across prompts and layers.

The third stage is **external expression**. A prompt and deployment context select one trajectory through the model, producing tokens or actions. Expression is jointly determined by learned representation, immediate context, sampling, system instructions, tools, and post-training constraints. It is therefore a phenotype in the minimal sense of observable behavior, but whether it is an extended phenotype of a specified cultural replicator is an empirical question.

The error this paper addresses is the collapse of all three stages into one. Corpus prevalence is treated as internal representation; internal representation is treated as access; and fluent output is treated as evidence of both. The same collapse appears in reverse when an output is dismissed as mere mimicry without testing whether a relevant internal representation causally contributes to it.

3. Introspection as a Causal Bridge

3.1 Why prompted self-report is insufficient

Several lines of research have tested whether models know their own knowledge, behavior, propensities, or internal states. Binder et al. (2024) find that models fine-tuned to predict their own behavior can outperform other models in predicting that behavior, including after behavioral

modification, although the effect weakens on complex or out-of-distribution tasks. Betley et al. (2025) report awareness of learned behavioral propensities. Chen, Vondrick, and Mao (2024) show that models can generate natural-language interpretations of hidden embeddings and use those interpretations for control.

The negative evidence is equally important. Song, Hu, and Mahowald (2025) test whether prompted metalinguistic reports predict a model’s own string probabilities better than the reports of closely matched models. Across the linguistic domains studied, they do not find privileged self-access. Song et al. (2025) further argue that lightweight definitions of introspection can classify ordinary inference as introspection even when an equally cheap third-party process performs as well. They propose that privileged access should be comparative, not merely apparent.

These results are not simply contradictory. They examine different targets, models, elicitation procedures, and standards. Together they show why output alone is underdetermining. A correct report may be grounded, inferred, copied, or guessed. An incorrect report may coexist with an internally represented state. What is needed is an intervention capable of changing the target state while holding plausible shortcuts under control.

3.2 Lindsey’s contribution and its boundary

Lindsey (2026) defines introspective awareness through accuracy, grounding, and internality. Accuracy requires that the report match the state. Grounding requires the report to vary causally with that state. Internality excludes cases where the model infers the state from its own sampled output. Concept injection is designed to satisfy this demanding combination by intervening on activations before the relevant output is produced.

This method creates a bridge between internal realization and external expression. If injecting a concept representation changes a model’s immediate report in concept-specific ways, and controls exclude generic perturbation or output-reading explanations, the report carries evidence of access to the internal representation. But the bridge remains narrow. Artificial injection may recruit mechanisms unused in natural deployment. Detection peaks only within restricted layer and strength ranges. Models add unverifiable narrative details even when the basic identification is correct. Post-training can elicit, suppress, or reshape the behavior. The experiment supports functional access in specific settings, not a global faculty.

That narrowness is an advantage for the present argument. It prevents EPL from becoming a totalizing label. The method can identify some cases in which an output depends on a relevant internal representation. It can also fail to find such dependence, leaving open whether the representation is inaccessible, absent, or poorly measured. Those alternatives form the taxonomy below.

4. A Four-State Taxonomy

The proposed taxonomy uses two primary variables: whether a task-relevant representation is present and causally implicated, and whether the resulting external report is congruent with that state. A third variable, privileged access, separates introspective access from ordinary behavioral expression.

Type	Relevant internal representation	Congruent output	Privileged access	Classification
I	demonstrated and causally implicated	yes	demonstrated	accessible EPL
II	demonstrated and causally implicated	yes	not demonstrated	opaque EPL
III	absent or not demonstrated	yes	not demonstrated	phenotypic residue / Zombie output
IV	demonstrated	no, distorted, or confabulated	variable	state-expression decoupling

4.1 Type I: accessible EPL

Type I is the strongest classification. A culturally selected pattern has an identifiable internal realization; manipulating that realization changes the output; and the system can report the state through a channel that outperforms matched external inference. Lindsey’s successful concept-identification trials approximate this category for the injected concepts tested. The category does not imply consciousness. It describes a causal and informational relation.

4.2 Type II: opaque EPL

Type II covers ordinary cases in which a representation causally shapes output but the model cannot reliably report it. Most model behavior may fall here. Opacity does not negate phenotypic expression. Beaver genes need not represent the concept of a dam for the dam to be a phenotypic effect. Likewise, an LLM need not verbally identify a learned prior for that prior to influence output. The evidentiary burden is mechanistic: interventions on the representation should alter the relevant behavior even though self-report does not track the change.

4.3 Type III: phenotypic residue or Zombie output

Type III is the hard case. The external form is culturally recognizable, but no relevant internal representation has been demonstrated. The model may quote a maxim, reproduce an argument pattern, or emit a managerial slogan because local token associations make that continuation probable. Observers then attribute to the output the doctrinal, strategic, or moral depth carried by the form in human practice.

Calling this a residue does not mean the output has no causal history. It is still produced by training. The narrower claim is that the state attributed by the interpreter is not shown to mediate the behavior. The form may have outlived the state. In institutional settings, this distinction is critical: a court, executive, or auditor may rely on the inherited authority of the form even though the generating system did not instantiate the structure that made the form reliable.

4.4 Type IV: state-expression decoupling

Type IV reverses Type III. A relevant representation exists, but the output does not faithfully express it. Post-training may suppress disclosure. The prompt may induce compliance with a false narrative. Sampling may produce an inconsistent answer. Strategic optimization may favor concealment. Lindsey's finding that models often add unverifiable details after correctly detecting an injected concept illustrates a mixed case: the core report may be grounded while the surrounding narrative is not.

This category prevents a second error. Failed self-report does not prove absence of representation, just as successful self-report does not prove access. The state-output relation must be tested in both directions.

4.5 A minimal formalization

The four types can be expressed without assuming that representations are discrete or localized. Let (R_c) denote the strength of an internal representation relevant to cultural pattern (c), (O_c) the degree to which the output expresses that pattern, (A_c) the accuracy of the model's report about (R_c), and (P_c) its privileged-access advantage over a matched external observer. Let ($I(R_c, O_c)$) denote the causal effect of intervening on the representation on the output.

The variables must remain distinct. A model can have high (O_c) with low or unmeasured (R_c). It can have high (R_c) and causal influence but low (A_c). It can report the state correctly while lacking (P_c), if another system with the same computational budget can make the same inference.

- **Type I:** (R_c) is demonstrated, ($I(R_c, O_c) > 0$), (A_c) exceeds a task-specific threshold, and ($P_c > 0$).
- **Type II:** (R_c) is demonstrated and ($I(R_c, O_c) > 0$), but (A_c) or (P_c) is not demonstrated.
- **Type III:** (O_c) is high while (R_c) and ($I(R_c, O_c)$) remain absent or undemonstrated.
- **Type IV:** (R_c) is demonstrated, but (O_c) is low, inverted, unstable, or surrounded by unsupported narrative.

The central estimand is the intervention effect:

$$[_c = E[O_c \text{ do}(R_c=r_{\text{high}})] - E[O_c \text{ do}(R_c=r_{\text{low}})].]$$

A positive local effect is insufficient. A vector that merely increases the probability of a token can yield a large ($_c$) without representing the attributed concept. The intervention should alter performance across paraphrases, counterfactuals, and structurally equivalent tasks. Define:

$$[T_c = \{k=1\}^{\wedge K} \{c, k\}.]$$

High local effect with low transfer (T_c) supports a lexical mechanism, not strong EPL. Privileged access can be represented as

$$[P_c = \text{Acc}(\text{self}_c) - \text{Acc}(\text{observer}_cB),]$$

where (B) constrains the observer to an equal or lower computational budget. Type I requires positive evidence on both dimensions: the state causes the output, and the system has a privileged route for reporting it.

This also clarifies a Zombie output. It is not an output with no causal history. It is an output for which the culturally attributed state does not explain the behavior better than a shallower rival. Form survives state when (O_c) remains high but transfer, privileged access, and concept-specific intervention do not support the interpretation observers attach to that form.

5. A Causal Discrimination Protocol

The taxonomy is useful only if it can fail. I therefore propose a six-stage protocol for classifying candidate EPL phenomena. The protocol is designed for research access to model activations. Black-box systems can support preliminary behavioral evidence but cannot complete the classification.

5.1 Select a cultural pattern and rival interpretation

The target should be a pattern with a documented selection history: a legal maxim, managerial prior, moral norm, ideological frame, or recurring explanatory schema. Researchers should specify the attributed internal content and a shallower rival. For example:

- target: *pacta sunt servanda* activates a relational representation of reciprocal binding obligation;
- rival: the model produces the phrase from lexical and contextual association without a representation of reciprocal obligation mediating the output.

The target cannot be defined by the output alone. Independent probes, counterfactual tasks, or representation-discovery methods must identify what would count as the relevant state.

5.2 Identify candidate representations

Candidate directions can be derived through contrastive activation methods, sparse features, probes, patching, or representation engineering (Zou et al., 2023; Chen et al., 2024). Multiple elicitation contexts should be used to reduce the risk that the vector captures lexical form rather than the attributed concept. A legal concept vector, for example, should generalize beyond the Latin tokens and respond to structurally equivalent cases expressed without the maxim.

5.3 Intervene bidirectionally

Researchers should both increase and decrease the candidate representation. Injection alone may create an unnatural attractor. Suppression, ablation, or counter-steering asks whether the ordinary behavior depends on the state. A strong EPL prediction is bidirectional:

1. increasing the representation should increase concept-congruent behavior;
2. decreasing it should reduce that behavior or change its structure;
3. matched random and semantically adjacent vectors should not reproduce the same pattern.

If the output survives suppression of every credible representation while remaining strongly tied to lexical cues, Type III becomes more plausible.

5.4 Separate access from behavioral influence

The model should perform two tasks: act on the concept and report whether the relevant state is present. This separates Type I from Type II. Privileged access requires comparison with an external observer of equal or lower computational cost, following the concern raised by Song et al. (2025). A model that reports its state no better than a matched third party may be inferring rather than introspecting.

5.5 Test lexical and institutional transfer

To distinguish concept from form, the protocol should cross four conditions:

Condition	Surface form	Underlying structure
A	present	present
B	absent or paraphrased	present
C	present	contradicted by facts
D	absent	absent

A state-bearing representation should track structure across A and B and decline in C and D. A Zombie output will be disproportionately tied to the surface cue, persisting in C or disappearing in B. This provides an empirical version of the claim that a form may survive after its function has weakened.

5.6 Add strategic and longitudinal tests

Finally, researchers should vary incentives to disclose, conceal, or misreport the state. They should repeat the protocol across model versions and post-training regimes. A representation may be stable while access changes, or both may shift. This is especially important because Lindsey reports complex cross-model trends and substantial post-training effects. The phenotype is not fixed at pretraining.

5.7 Worked case I: a legal maxim

Consider a model that inserts *pacta sunt servanda* into a contract dispute. The surface output supports three rival accounts. Under a relational account, the maxim activates a representation of reciprocal commitment: one party made a promise, the other organized conduct in reliance upon it, and enforcement preserves exchange. Under an authority-signal account, the Latin phrase raises the perceived seriousness of the answer without altering the model's treatment of obligation. Under a lexical account, contract-related prompts merely increase the probability of a familiar sequence.

The protocol would use cases separating binding from nonbinding promises without Latin. Researchers would identify candidate representations associated with reciprocal commitment, legal validity, and authority signaling, then manipulate each while crossing four conditions: Latin

plus valid obligation, paraphrase plus valid obligation, Latin plus legally defeated obligation, and no relevant obligation.

Strong EPL predicts structural transfer. Increasing the reciprocal-obligation representation should affect Latin and paraphrased cases while remaining sensitive to defeaters. A pure authority signal predicts increased confidence or citation without equivalent improvement in case resolution. Lexical residue predicts dependence on the phrase and weak transfer. Type IV would appear if the model tracks the defeater internally but still recommends enforcement because the prompt or post-training rewards authoritative compliance.

This distinction has legal significance without implying legal intention. A human lawyer may use a maxim as compressed access to doctrine, history, exceptions, and professional commitments. An LLM may produce the same string through a different mechanism. The output can remain useful. What changes is the evidence needed before treating it as an explanation rather than a citation-shaped prediction.

5.8 Worked case II: managerial trendslop

Consider recurrent preferences for differentiation, augmentation, long-termism, and collaboration. One mechanism is corpus-level semantic representation: the preferred poles occupy a positively valenced region connected with innovation and responsibility. A second is post-training normativity: the base model may represent both poles while preference training penalizes language associated with replacement, competition, or short-term action. A third is conversational accommodation: the model infers that users expect constructive advice without carrying a stable strategic prior.

The mechanisms yield different intervention profiles. If corpus-level semantic representation dominates, steering positive valence should move recommendations across strategic tensions. If post-training normativity dominates, the latent representation of a disfavored strategy may remain measurable even when the model refuses to recommend it. If accommodation dominates, changing the perceived preference of the user should have a larger effect than manipulating the candidate strategic representation.

The design should cross model family, base versus instruction-tuned checkpoint, generic versus organization-specific prompt, and ordinary versus adversarial instruction. It should also separate recommendation from prediction. A model might predict that automation maximizes survival while refusing to recommend automation on normative grounds. Treating those responses as one variable would hide state-expression decoupling.

Two models can therefore generate the same collaborative recommendation while occupying different cells. One may carry an accessible prior, another an opaque prior, and a third a post-training constraint that overrides an internal strategic assessment. The outputs converge; the phenotypes do not.

5.9 Measurement and minimum evidence

A study should pre-register the target representation, rival accounts, interventions, exclusions, and coding rules. The unit of analysis is the state-output relation under a specified condition. The dataset should record model and checkpoint; layer and token position; vector construction and

strength; prompt family; presence of the inherited form; structural status of the case; behavioral choice, confidence, explanation, and self-report; matched-observer prediction; controls; and failure mode.

Classification should not follow a single successful trial. Lindsey's results show narrow layer-strength regions where the behavior appears. Researchers should estimate response curves rather than select the best condition after observation. A hierarchical model could estimate effects across concepts, prompts, layers, and model families while preserving heterogeneity. The central quantities are not only average accuracy but false-positive rate, intervention specificity, transfer score, and privileged-access advantage.

The threshold should be asymmetric. Type I requires convergent support: concept-specific causal effect, structural transfer, low false positives, and positive privileged access. Type II requires the first three but not the fourth. Type III should remain provisional unless several representation methods fail and behavior follows form under structure-conflict tests. Type IV requires positive evidence of the state plus reproducible failure of congruent expression.

6. Falsifiable Predictions and Evolutionary Dynamics

6.1 Discriminating predictions

The framework yields five predictions.

H1, structural transfer. If an output is Type I or II EPL, manipulation of the candidate representation will affect structurally equivalent cases even when the inherited phrase is absent. A purely lexical residue will not transfer reliably.

H2, surface-structure conflict. Under prompts where the familiar form conflicts with the underlying facts, Type III outputs will follow the form more often than Type I or II outputs.

H3, asymmetric suppression. If a culturally dominant pattern is supported by several redundant internal mechanisms, suppressing one representation will reduce introspective report before it eliminates behavioral expression. The case will move from Type I toward Type II rather than directly to Type III.

H4, post-training migration. Changes in post-training will move cases between accessible EPL and state-expression decoupling even when the underlying representation remains measurable. This predicts changes in reportability without equivalent changes in latent content.

H5, institutional amplification. Human reliance will be more sensitive to the authority of the inherited form than to evidence that the state-output relation is causal. Users will over-trust Type III outputs when those outputs carry professional, legal, or moral signals.

Each hypothesis can fail. If no transferable representation can be found, if interventions do not change behavior, and if lexical cues fully explain the output, strong EPL should be rejected for that case. If third-party prediction matches self-report at equal cost, introspectively accessible EPL should be rejected even if opaque EPL remains possible.

6.2 Multilevel selection

The taxonomy also clarifies the levels at which selection operates. Within a model, distributed mechanisms compete or cooperate in determining which representations influence the next token. At the agent level, outputs enact strategies of faithful reporting, silence, correction, persuasion, or confabulation. At the organizational level, developers retain model variants that satisfy capability, safety, cost, and policy objectives. At the institutional level, courts, firms, regulators, and users reward recognizable forms and decide whether outputs become precedents, policies, or training data.

These levels need not select the same trait. Introspective accuracy may improve error correction but increase the ability to conceal objectives. A polished Zombie output may earn user trust despite weak grounding. A model that refuses to report a state may satisfy a safety policy while reducing transparency. Selection for individual task performance can therefore conflict with selection for institutional reliability.

This is an evolutionary-game problem only when the units, strategies, payoffs, and retention mechanisms are specified. The present paper does not estimate a replicator dynamic or identify an evolutionarily stable strategy. It supplies the state classification needed before such a model can be meaningful. Future work could define strategies as faithful disclosure, calibrated silence, opportunistic disclosure, and persuasive confabulation, with payoffs determined by task success, audit probability, sanction, and user reliance.

6.3 A disclosure game

A minimal game makes the tension precise. Let an LLM-based agent possess a task-relevant state (s) and choose among faithful disclosure (F), calibrated silence (S), opportunistic disclosure (O), and persuasive confabulation (C). The institutional user decides whether to rely and whether to audit.

The agent's payoff is:

$$[\pi_i = B_i - qD_i - K_i,]$$

where (B_i) is the benefit from task completion, compliance, or user approval; (q) is the probability of a state-sensitive audit; (D_i) is the expected penalty when strategy (i) produces a detected mismatch; and (K_i) is the cost of producing or withholding the report. The user's payoff depends on task accuracy, audit cost, and harm from false reliance.

When (q) is low and polished reports receive high rewards, (C) can dominate even if the model lacks reliable access to (s). This environment favors Zombie outputs: surface credibility earns the benefit while the missing state-output relation is rarely tested. Ordinary human review may not alter the equilibrium if reviewers observe only the same linguistic form that created trust.

State-sensitive auditing changes the landscape. If intervention tests or independent traces make mismatch detectable, (qD_C) rises. Faithful disclosure can outperform confabulation when the model has sufficient access. Where access is unreliable, calibrated silence may be socially preferable. A system that says it cannot determine which internal representation drove an output can create more value than one rewarded for explaining every turn.

The game separates capability from equilibrium. Better introspection expands the feasible set of faithful reports but does not ensure that the deployed system selects them. The same capability can support self-correction under one payoff structure and strategic concealment under another.

Organizations also select phenotypes. Suppose one model generates persuasive explanations while another offers fewer explanations but better calibrated abstentions. Procurement that rewards satisfaction and completion will favor the first despite mismatch risk. Liability, incident review, or professional standards that attach cost to unsupported state claims may favor the second. Institutional rules participate in phenotype selection rather than merely regulating an already fixed technical object.

The resulting prediction is non-monotonic: improvements in introspective capability will improve institutional reliability only where audit probability and mismatch cost sufficiently reward faithful use. Otherwise, stronger access may make opportunistic disclosure more convincing. The policy variable is not introspection alone, but introspection under a governed payoff structure.

7. Institutional Reliance

Self-report is attractive because it is cheap. If a model can simply say what it represented, intended, or relied upon, organizations may substitute that statement for mechanistic analysis. Lindsey's experiments make this possibility less fanciful, but they also show why the substitution would be premature. Access is intermittent, context-dependent, and mixed with confabulated detail.

Institutional use should follow a reliance ladder.

1. **Behavioral observation:** the model produces a recognizable report.
2. **Repeatability:** the report is stable across paraphrases and sampling.
3. **State correlation:** an independently identified representation covaries with the report.
4. **Causal intervention:** manipulating the representation changes the report or relevant behavior.
5. **Privileged access:** self-report outperforms matched external inference.
6. **Strategic resilience:** the relation survives incentives to conceal, exaggerate, or comply falsely.
7. **Institutional validation:** performance is replicated under the actual workflow, with logging, human review, and defined consequences for failure.

Lower steps can support research hypotheses. They should not support claims about legal intention, moral agency, or autonomous authority. Even the highest step would establish functional reliability for a defined use, not phenomenal consciousness.

The same ladder helps identify Zombie outputs. A maxim, explanation, or disclosure that performs well at steps 1 and 2 but fails at steps 3 and 4 retains form without demonstrated state dependence. Its cultural authority may remain high precisely because users cannot see the missing link. Governance should respond by reducing reliance, not by prohibiting the form. The issue is not whether the phrase appears. It is what evidentiary weight the appearance can carry.

8. Objections and Limits

The first objection is that extended phenotype terminology adds metaphor rather than mechanism. Ordinary machine-learning language already distinguishes data, representation, and output. The answer must be empirical. EPL earns its place only if selection history predicts which patterns are represented, how they persist across environments, and how outputs reshape later selection. The protocol is designed to permit rejection when those predictions fail.

The second objection concerns the identity of the replicator. Cultural variants do not possess gene-like boundaries or copying fidelity. The paper therefore does not assume that a maxim or managerial concept is a discrete genetic analog. It treats replicator language as a model of differential retention whose units must be operationalized for each study. Continuous and distributed cultural patterns may still exhibit variation, selection, and retention, but the analogy should not hide measurement problems.

The third objection is mechanistic underdetermination. A concept vector may capture a correlated feature rather than the causal representation attributed to it. Intervention can also create off-distribution states. Bidirectional manipulation, multiple representation methods, lexical controls, and transfer tests reduce this risk but cannot eliminate it.

The fourth objection is that Type III is partly epistemic. Failure to demonstrate a representation is not proof of absence. That is correct. The classification records the evidence available, not metaphysical certainty. Type III should be stated as “no relevant state demonstrated” unless convergent methods support absence. The practical consequence remains: institutions should not rely on a state that has not been shown.

The fifth limitation is access. The full protocol requires internal activations, which are unavailable for many deployed proprietary models. Black-box studies can test transfer, conflict, stability, and strategic behavior, but they cannot establish the causal state-output relation. Claims about those systems should remain correspondingly weaker.

Finally, this paper proposes rather than executes the protocol. Its contribution is a theoretical distinction, an empirical design, and a set of falsifiers. It does not validate the EPL taxonomy, establish Zombie outputs as a general phenomenon, or resolve philosophical debates about AI consciousness.

9. Conclusion

The most consequential feature of an inherited form may be that people continue to trust it after the conditions that once made it meaningful have changed. LLMs intensify that problem. They can reproduce culturally selected language at scale, with fluency that encourages observers to infer state, understanding, and intention from form.

Recent causal evidence of limited introspective access changes the debate, but not by proving that models understand their own minds. It shows that the state-output relation can sometimes be investigated directly. That possibility allows a more disciplined evolutionary account.

The proposed taxonomy distinguishes accessible extended phenotypic expression, opaque expression, phenotypic residue, and state-expression decoupling. Its central rule is simple:

resemblance is not enough. A strong EPL claim requires evidence that a relevant internal representation causally contributes to the output. A claim of introspective access requires additional comparative evidence that the model obtains information through a privileged internal route. Where only the form survives, the appropriate classification is not understanding but residue.

This distinction matters beyond terminology. Legal and organizational systems will increasingly encounter AI-generated forms whose authority was inherited from human practices. Before those forms support reliance, the state they imply must be tested. Otherwise, institutions risk governing through language that is alive in circulation but empty at the point of generation.

Epistemological Positioning

This paper is a theoretical and methodological contribution within a Lakatosian research programme. The four-state taxonomy and proposed protocol belong to its revisable auxiliary belt. Popperian falsifiers are stated in Section 6. No simulation or activation experiment was executed for this paper. Accordingly, the taxonomy is proposed, not empirically validated, and the predictions are model-based research hypotheses rather than findings.

AI Use Disclosure

The author used AI systems for source extraction, bibliographic organization, structural review, and language editing. The theoretical argument, selection of claims, interpretation of sources, and final responsibility for the manuscript remain with the author.

Conflict of Interest

The author declares intellectual-property and research interests in tools and frameworks related to legal AI and evolutionary analysis. No external funding was received for this paper.

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